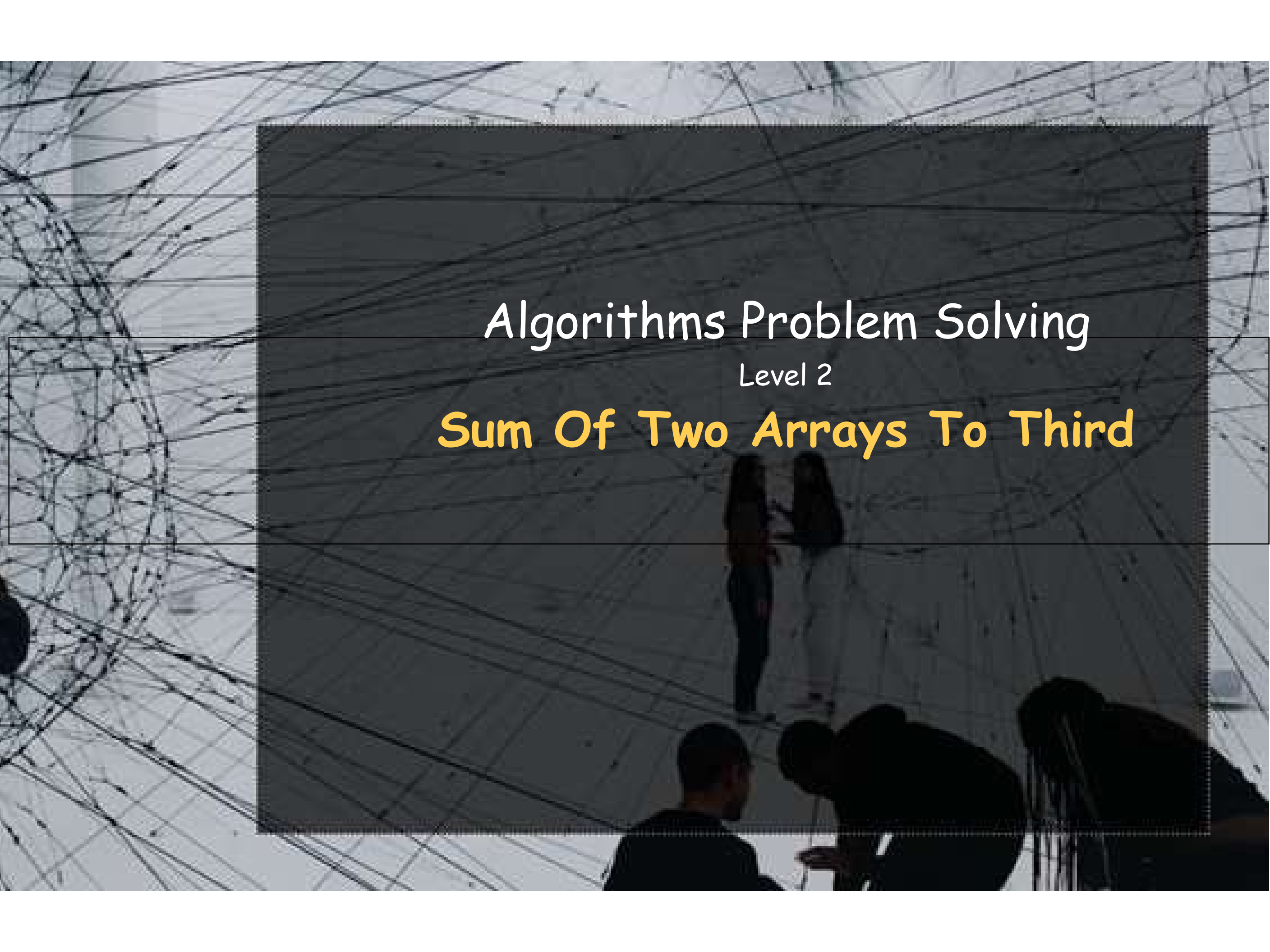


Algorithms Problem Solving Level 2

Sum Of Two Arrays To Third



Algorithms Problem Solving

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Sum Of Two Arrays To Third

Definition

- ◆ This program fills two arrays with random numbers from `1` to `100`.
- ◆ It adds the corresponding elements of the two arrays.
- ◆ The results are stored in a third array.
- ◆ Finally, it prints all three arrays.

Explanation

- ◆ ``FillArrayWithRandomNumber()`` fills an array with random numbers from ``1`` to ``100``.
- ◆ ``SumOfTwoArrays()`` adds the elements at the same index from ``arr1`` and ``arr2``.
- ◆ The result is stored in ``arrSum``.
- ◆ For example:
$$\text{arrSum}[0] = \text{arr1}[0] + \text{arr2}[0];$$
- ◆ If ``arr1[0]`` is ``30`` and ``arr2[0]`` is ``48``, then:
$$30 + 48 = 78$$
- ◆ The process is repeated for every element.

Steps in main()

1. Ask the user for the number of elements.
2. Fill `arr1` with random numbers from `1` to `100`.
3. Fill `arr2` with random numbers from `1` to `100`.
4. Loop through both arrays using the same index.
5. Add `arr1[i]` and `arr2[i]`.
6. Store the result in `arrSum[i]`.
7. Print `arr1`.
8. Print `arr2`.
9. Print `arrSum`.

💡 Example

◆ `Array 1`:

30 95 2 18 63 91 70 15 54 14

◆ `Array 2`:

48 81 97 86 58 4 72 35 71 45

◆ The addition is performed element by element:

$$30 + 48 = 78$$

$$95 + 81 = 176$$

$$2 + 97 = 99$$

$$18 + 86 = 104$$

$$63 + 58 = 121$$

$$91 + 4 = 95$$

$$70 + 72 = 142$$

$$15 + 35 = 50$$

$$54 + 71 = 125$$

$$14 + 45 = 59$$

➡ Result

Array 1 Elements: 30 95 2 18 63 91 70 15 54 14

Array 2 Elements: 48 81 97 86 58 4 72 35 71 45

Sum of array1 and array2 Elements: 78 176 99 104 121 95 142 50 125 59

Note

- ◆ The important statement is:
`arrSum[i] = arr1[i] + arr2[i];`
- ◆ The same index `i` is used in all three arrays.
- ◆ `arr1[i]` and `arr2[i]` are added together and the result goes into `arrSum[i]`.
- ◆ All three arrays have the same length in this program.
- ◆ `counter` inside `SumOfTwoArrays()` is declared but never used:
`int counter = 0;`
- ◆ It can simply be removed.

⊘ Warning

- ◆ The program should make sure `arrayLength` does not exceed `100`.
- ◆ `ReadArrayElements()` is defined but not used.
- ◆ `cmath` and `corecrt_math_defines.h` are unnecessary for this program.
- ◆ `FillArrayWithRandomNumber()` assumes that `arrLength` has already been set correctly.
- ◆ Since `srand((unsigned)time(NULL))` is used, the program needs the appropriate time/random declarations; normally `<ctime>` and `<cstdlib>` should be included explicitly.

Comparison

Comparison

`arr1` and `arr2`

- ✓ Contain the original random numbers.
- ✓ Both arrays are used as input for the calculation.
- ✓ Example: `arr1[0] = 30`, `arr2[0] = 48`.

`arrSum`

- ✓ Stores the result of adding the two arrays.
- ✓ Each element is calculated as `arr1[i] + arr2[i]`.
- ✓ Example: `30 + 48` → `78`.

✦ Summary

- ✓ Create two arrays with random numbers.
- ✓ Use the same index to access corresponding elements.
- ✓ Add `arr1[i]` and `arr2[i]`.
- ✓ Store the result in `arrSum[i]`.
- ✓ The third array contains the sum of the two original arrays.

Thank you 😊